Part A - Literature Review

Title	The determinants of investment fraud: A machine learning and artificial intelligence approach
Harvard reference	Lokanan, M. (2022) 'The determinants of investment fraud: A machine learning and artificial intelligence approach', *Frontiers in Big Data*, 5, p. 961039. Available at: https://doi.org/10.3389/fdata.2022.961039 (Accessed: 22 September 2026).
Link	https://www.frontiersin.org/journals/big-data/articles/10.3389/fdata.2022.961039/full
Modelling	K-Nearest Neighbors (KNN), Random Forest, Gradient Boosting, Artificial Neural Network (ANN)
Preprocessing techniques	- Median was imputed into missing numerical data, and mode was imputed into missing categorical data. - Dataset was scaled using Standard Scaler - Categorical data was encoded into binary features - SMOTE +ENN
Datasets used	Data from the IIRCOC hearing panel between June 2008 and December 2019
Key takeaways	- The dataset is heavily imbalanced because fraud is a rare occurrence. Due to this we have to address this class imbalance either by oversampling or undersampling to avoid bias toward the majority class - Most of the victims of investment fraud are aiming to liquidate their fixed assets. These are the transactions and insights we may need to look out for. - Less experienced brokers are more likely to defraud clients than more experienced brokers. Broker tenure is a contributing factor in Fraud and may impact our results

Title	Fraud Detection in Financial Transactions: A Unified Deep Learning Approach
Harvard reference	Mazumder, M.T.R., Shourov, M.S.H., Rasul, I., Akter, S. and Miah, M.K. (2025) 'Fraud Detection in Financial Transactions: A Unified Deep Learning Approach', *Journal of Economics, Finance and Accounting Studies*, 7(2), pp. 184–194. Available at: https://doi.org/10.32996/jefas.2025.7.2.16 (Accessed: 22 September 2026).
Link	https://alkindipublisher.com/index.php/jefas/article/view/9167/7853
Modelling	Hybrid combined model which uses a Convolutional Neural Network, a Gated Recurrent Unit and an Attention Mechanism. - CNN - GRU - CNN-GRU - CNN-GRU-Attention
Preprocessing techniques	- Scaling with Min-Max normalization for Amount and Time Features - SMOTE - Test Train split of 70% training/15% validation /15% test
Datasets used	European Credit Card Fraud Detection Dataset
Key takeaways	- A combined model has significant advantages as it takes the strengths of each of the individual models into consideration. However it is unintuitive and may be difficult to explain to stakeholders. - The data used was not retail investment fraud data however the evaluation principles still apply as it is still transactional data. The data is anonymised for confidentiality purposes. - The model could have benefitted greatly from additional data sources. Gaining as much data as possible will produce better results

Title	Credit Card Fraud Detection using Machine Learning Algorithms
Harvard reference	Dornadula, V.N. and Geetha, S. (2019) 'Credit card fraud detection using machine learning algorithms', *Procedia Computer Science*, 165, pp. 631–641. Available at: https://doi.org/10.1016/j.procs.2019.12.266 (Accessed: 23 September 2026).
Link	https://www.sciencedirect.com/science/article/pii/S18770509 2030065X
Modelling	- Local Outlier Factor - Isolation Forest - Support Vector Machine - Logistic Regression - Decision Tree - Random Forest
Preprocessin g techniques	- SMOTE
Datasets used	European Credit Card Fraud Detection Dataset
Key takeaways	- MCC is useful when evaluating imbalanced datasets. I will be incorporating this into my model evaluation - One class SVM can effectively handle class imbalance by training on the majority class and detecting anomalies. It may be a model to consider using. - This paper focuses on grouping each specific customer into a customer behaviour cluster and evaluating subsequent transactions on their current cluster. This is an effective way to determine fraud since our models can evaluate anomaly behaviour based on past client behaviour.

Dataset which will be used in Part B

Name	European Credit Card Fraud Detection Dataset
Harvard reference	Machine Learning Group ULB (2018) *Credit Card Fraud Detection* [Data set]. Available at: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud (Accessed: 22 September 2026).
Link	https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud
Number of instances	284,807
Number of feature columns	31
Number of classes (if applicable)	2
Target range (if applicable)	0-1
Origin	**The dataset contains credit card transactions of European cardholders in September 2013. These transactions occurred over 2 days and has 492 Frauds out of 284,807 transactions. The data was transformed by PCA and contains only numerical values. PCA was performed to keep the confidentiality of the data. For the same reason we do not have access to the feature titles. We only have access to 'Amount','Time' and 'Class'**